6
Probability Spaces

6.1 Kolmogorov's probability model

Probability theory provides a mathematical model for random phenomena, i.e., those involving uncertainty. First one identifies the set Ω of possible outcomes of (random) experiment associated with the phenomenon. This set Ω is called the *sample space*, and an individual element ω of Ω is called a *sample point*. Even though the outcome is not predictable ahead of time, one is interested in the "chances" of some particular statement to be valid for the resulting outcome. The set of ω's for which a given statement is valid is called an *event*. Thus, an event is a subset of Ω. One then identifies a class $\mathcal{F}$ of events, i.e., a class $\mathcal{F}$ of subsets of Ω (not necessarily all of $\mathcal{P}(\Omega)$, the power set of Ω), and then a set function P on $\mathcal{F}$ such that for A in $\mathcal{F}$, $P(A)$ represents the "chance" of the event A happening. Thus, it is reasonable to impose the following conditions on $\mathcal{F}$ and P:

(i) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$
(i.e., if one can define the probability of an event A, then the probability of A not happening is also well defined).

(ii) $A_1, A_2 \in \mathcal{F} \Rightarrow A_1 \cup A_2 \in \mathcal{F}$
(i.e., if one can define the probabilities of A_1 and A_2, then the probability of at least one of A_1 or A_2 happening is also well defined).

(iii) for all A in $\mathcal{F}$, $0 \leq P(A) \leq 1$, $P(\emptyset) = 0$, and $P(\Omega) = 1$.

(iv) $A_1,\, A_2 \in \mathcal{F},\, A_1 \cap A_2 = \emptyset \Rightarrow P(A_1 \cup A_2) = P(A_1) + P(A_2)$ (i.e., if A_1 and A_2 are mutually exclusive events, then the probability of at least one of the two happening is simply the sum of the probabilities).

The above conditions imply that $\mathcal{F}$ is an *algebra* and P is a *finitely additive* set function. Next, as explained in Section 1.2, it is natural to require that $\mathcal{F}$ be closed under monotone increasing unions and P be monotone continuous from below. That is, if $\{A_n\}_{n\geq 1}$ is a sequence of events in $\mathcal{F}$ such that A_n implies A_{n+1} (i.e., $A_n \subset A_{n+1}$) for all $n \geq 1$, then the probability of at least one of the A_n's happening is well defined and is the limit of the corresponding probabilities. In other words, the following conditions on $\mathcal{F}$ and P must hold in addition to (i)–(iv) above:

(v) $A_n \in \mathcal{F},\ A_n \subset A_{n+1}$ for all $n = 1, 2, \ldots \quad \Rightarrow \bigcup_{n\geq 1} A_n, \in \mathcal{F}$ and $P(A_n) \uparrow P(\bigcup_{n\geq 1} A_n)$.

As noted in Section 1.2, conditions (i)–(v) imply that $(\Omega, \mathcal{F}, P)$ is a *measure space*, i.e., $\mathcal{F}$ is a σ-algebra and P is a measure on $\mathcal{F}$ with $P(\Omega) = 1$. That is, $(\Omega, \mathcal{F}, P)$ is a *probability space*. This is known as *Kolmogorov's probability model* for random phenomena (see Kolmogorov (1956), Parthasarathy (2005)). Here are some examples.

Example 6.1.1: (*Finite sample spaces*). Let $\Omega \equiv \{\omega_1, \omega_2, \ldots, \omega_k\}, 1 \leq k < \infty$, $\mathcal{F} \equiv \mathcal{P}(\Omega)$, the power set of Ω and $P(A) = \sum_{i=1}^k p_i I_A(\omega_i)$ where $\{p_i\}_{i=1}^k$ are such that $p_i \geq 0$ and $\sum_{i=1}^k p_i = 1$. This is a probability model for random experiments with finitely many possible outcomes.

An important application of this probability model is finite population sampling. Let $\{U_1, U_2, \ldots, U_N\}$ be a finite population of N units or objects. These could be individuals in a city, counties in a state, etc. In a typical sample survey procedure, one chooses a subset of size n $(1 \leq n \leq N)$ from this population. Let Ω denote the collection of all possible subsets of size n. Here $k = \binom{N}{n}$, each ω_i is a sample of size n and p_i is the selection probability of ω_i. The assignment of $\{p_i\}_{i=1}^k$ is determined by a given *sampling scheme*. For example, in simple random sampling without replacement, $p_i = \frac{1}{k}$ for $i = 1, 2, \ldots, k$.

Other examples include coin tossing, rolling of dice, bridge hands, and acceptance sampling in statistical quality control (Feller (1968)).

Example 6.1.2: (*Countably infinite sample spaces*). Let $\Omega \equiv \{\omega_1, \omega_2, \ldots\}$ be a countable set, $\mathcal{F} = \mathcal{P}(\Omega)$, and $P(A) \equiv \sum_{i=1}^\infty p_i I_A(\omega_i)$ where $\{p_i\}_{i=1}^\infty$ satisfy $p_i \geq 0$ and $\sum_{i=1}^\infty p_i = 1$. It is easy to verify that $(\Omega, \mathcal{F}, P)$ is a probability space. This is a probability model for random experiments with countably infinite number of outcomes. For example, the experiment of tossing a coin until a "head" is produced leads to such a probability space.

Example 6.1.3: (*Uncountable sample spaces*).

(a) (*Random variables*). Let $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$, $P = \mu_F$, the Lebesgue-Stieltjes measure corresponding to a cdf F, i.e., corresponding to a function $F : \mathbb{R} \to \mathbb{R}$ that is nondecreasing, right continuous, and satisfies $F(-\infty) = 0,\ F(+\infty) = 1$. See Section 1.3. This serves as a model for a single random variable X.

(b) (*Random vectors*). Let $\Omega = \mathbb{R}^k$, $\mathcal{F} = \mathcal{B}(\mathbb{R}^k)$, $P = \mu_F$, the Lebesgue-Stieltjes measure corresponding to a (multidimensional) cdf F on $\mathbb{R}^k$ where $k \in \mathbb{N}$. See Section 1.3. This is a model for a random vector $(X_1, X_2, \ldots, X_k)$.

(c) (*Random sequences*). Let $\Omega = \mathbb{R}^\infty \equiv \mathbb{R} \times \mathbb{R} \times \ldots$ be the set of all sequences $\{x_n\}_{n \geq 1}$ of real numbers. Let $\mathcal{C}$ be the class of all finite dimensional sets of the form $A \times \mathbb{R} \times \mathbb{R} \times \ldots$, where $A \in \mathcal{B}(\mathbb{R}^k)$ for some $1 \leq k < \infty$. Let $\mathcal{F}$ be the σ-algebra generated by $\mathcal{C}$. For each $1 \leq k < \infty$, let μ_k be a probability measure on $\mathcal{B}(\mathbb{R}^k)$ such that $\mu_{k+1}(A \times \mathbb{R}) = \mu_k(A)$ for all $A \in \mathcal{B}(\mathbb{R}^k)$. Then there exists a probability measure μ on $\mathcal{F}$ such that $\mu(A \times \mathbb{R} \times \mathbb{R} \times \ldots) = \mu_k(A)$ if $A \in \mathcal{B}(\mathbb{R}^k)$. (This will be shown later as a special case of the Kolmogorov's consistency theorem in Section 6.3.) This will be a model for a sequence $\{X_n\}_{n \geq 1}$ of random variables such that for each k, $1 \leq k < \infty$, the distribution of $(X_1, X_2, \ldots, X_k)$ is μ_k.

6.2 Random variables and random vectors

Recall the following definitions introduced earlier in Sections 2.1 and 2.2.

Definition 6.2.1: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X : \Omega \to \mathbb{R}$ be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable, that is, $X^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{B}(\mathbb{R})$. Then, X is called a *random variable* on $(\Omega, \mathcal{F}, P)$.

Recall that $X : \Omega \to \mathbb{R}$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable iff for all $x \in \mathbb{R}$, $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$.

Definition 6.2.2: Let X be a random variable on $(\Omega, \mathcal{F}, P)$. Let

$$F_X(x) \equiv P(\{\omega : X(\omega) \leq x\}), \quad x \in \mathbb{R}. \tag{2.1}$$

Then $F_X(\cdot)$ is called the *cumulative distribution function* (cdf) of X.

Definition 6.2.3: Let X be a random variable on $(\Omega, \mathcal{F}, P)$. Let

$$P_X(A) \equiv P(X^{-1}(A)) \quad \text{for all} \quad A \in \mathcal{B}(\mathbb{R}). \tag{2.2}$$

Then the probability measure P_X is called the *probability distribution* of X.

Note that P_X is the measure induced by X on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ under P and that the Lebesgue-Stieltjes measure μ_{F_X} on $\mathcal{B}(\mathbb{R})$ corresponding with the cdf F_X of X is the same as P_X.

Definition 6.2.4: Let $(\Omega, \mathcal{F}, P)$ be a probability space, $k \in \mathbb{N}$ and $X : \Omega \to \mathbb{R}^k$ be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^k) \rangle$-measurable, i.e., $X^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{B}(\mathbb{R}^k)$. Then X is called a (k-dimensional) *random vector* on $(\Omega, \mathcal{F}, P)$.

Let $X = (X_1, X_2, \ldots, X_k)$ be a random vector with components X_i, $i = 1, 2, \ldots, k$. Then each X_i is a random variable on $(\Omega, \mathcal{F}, P)$. This follows from the fact that the coordinate projection maps from $\mathbb{R}^k$ to $\mathbb{R}$, given by

$$\pi_i(x_1, x_2, \ldots, x_k) \equiv x_i, \ 1 \le i \le k$$

are continuous and hence, are Borel measurable. Conversely, if for $1 \le i \le k$, X_i is a random variable on $(\Omega, \mathcal{F}, P)$, then $X = (X_1, X_2, \ldots, X_k)$ is a random vector (cf. Proposition 2.1.3).

Definition 6.2.5: Let X be a k-dimensional random vector on $(\Omega, \mathcal{F}, P)$ for some $k \in \mathbb{N}$. Let

$$F_X(x) \equiv P(\{\omega : X_1(\omega) \le x_1, X_2(\omega) \le x_2, \ldots, X_k(\omega) \le x_k\}) \qquad (2.3)$$

for $x = (x_1, x_2, \ldots, x_k) \in \mathbb{R}^k$. Then $F_X(\cdot)$ is called the *joint cumulative distribution function* (joint cdf) of the random vector X.

Definition 6.2.6: Let X be a k-dimensional random vector on $(\Omega, \mathcal{F}, P)$ for some $k \in \mathbb{N}$. Let

$$P_X(A) = P(X^{-1}(A)) \quad \text{for all} \quad A \in \mathcal{B}(\mathbb{R}^k). \qquad (2.4)$$

The probability measure P_X is called the *(joint) probability distribution* of X.

As in the case $k = 1$, the Lebesgue-Stieltjes measure μ_{F_X} on $\mathcal{B}(\mathbb{R}^k)$ corresponding to the joint cdf F_X is the same as P_X.

Next, let $X = (X_1, X_2, \ldots, X_k)$ be a random vector. Let $Y = (X_{i_1}, X_{i_2}, \ldots, X_{i_r})$ for some $1 \le i_1 < i_2 < \ldots < i_r \le k$ and some $1 \le r \le k$. Then, Y is also a random vector. Further, the joint cdf of Y can be obtained from F_X by setting the components x_j, $j \notin \{i_1, i_2, \ldots, i_r\}$ equal to ∞. Similarly, the probability distribution P_Y can be obtained from P_X as an induced measure from the projection map $\pi(x) = (x_{i_1}, x_{i_2}, \ldots, x_{i_r})$, $x \in \mathbb{R}^k$. For example, if $(i_1, i_2, \ldots, i_r) = (1, 2, \ldots, r)$, $r \le k$, then

$$F_Y(y_1, \ldots, y_r) = F_X(y_1, \ldots, y_r, \infty, \ldots, \infty), \ (y_1, \ldots, y_r) \in \mathbb{R}^r$$

and

$$P_Y(A) = P_X(A \times \mathbb{R}^{(k-r)}), \ A \in \mathcal{B}(\mathbb{R}^r).$$

Definition 6.2.7: Let $X = (X_1, X_2, \ldots, X_k)$ be a random vector on $(\Omega, \mathcal{F}, P)$. Then, for each $i = 1, \ldots, k$, the cdf F_{X_i} and the probability distribution P_{X_i} of the random variable X_i are called the *marginal* cdf and the *marginal probability distribution* of X_i, respectively.

It is clear that the distribution of X determines the marginal distribution P_{X_i} of X_i for all $i = 1, 2, \ldots, k$. However, the marginal distributions $\{P_{X_i} : i = 1, 2, \ldots, k\}$ do not uniquely determine the joint distribution P_X, without additional conditions, such as independence (see Problem 6.1).

Definition 6.2.8: Let X be a random variable on $(\Omega, \mathcal{F}, P)$. The *expected value* of X, denoted by EX or $E(X)$, is defined as

$$EX = \int_\Omega X dP, \tag{2.5}$$

provided the integral is well defined. That is, at least one of the two quantities $\int X^+ dP$ and $\int X^- dP$ is finite.

If X is a random variable on $(\Omega, \mathcal{F}, P)$ and $h : \mathbb{R} \to \mathbb{R}$ is Borel measurable, then $Y = h(X)$ is also a random variable on $(\Omega, \mathcal{F}, P)$. The expected value of Y may be computed as follows.

Proposition 6.2.1: (*The change of variable formula*). *Let X be a random variable on $(\Omega, \mathcal{F}, P)$ and $h : \mathbb{R} \to \mathbb{R}$ be Borel measurable. Let $Y = h(X)$. Then*

(i) $\displaystyle\int_\Omega |Y| dP = \int_\mathbb{R} |h(x)| P_X(dx) = \int_\mathbb{R} |y| P_Y(dy).$

(ii) If $\int_\Omega |Y| dP < \infty$, *then*

$$\int_\Omega Y dP = \int_\mathbb{R} h(x) P_X(dx) = \int_\mathbb{R} y P_Y(dy). \tag{2.6}$$

Proof: If $h = I_A$ for A in $\mathcal{B}(\mathbb{R})$, the proposition follows from the definition of P_X. By linearity, this extends to a nonnegative and simple function h and by the MCT, to any nonnegative measurable h, and hence to any measurable h. $\square$

Remark 6.2.1: Proposition 6.2.1 shows that the expectation of Y can be computed in three different ways, i.e., by integrating Y with respect to P on Ω or by integrating $h(x)$ on $\mathbb{R}$ with respect to the probability distribution P_X of the random variable X or by integrating y on $\mathbb{R}$ with respect to the probability distribution P_Y of the random variable Y.

Remark 6.2.2: If the function h is nonnegative, then the relation $EY = \int_\mathbb{R} h(x) P_X(dx)$ is valid even if $EY = \infty$.

Definition 6.2.9: For any positive integer n, the nth *moment* μ_n of a random variable X is defined by

$$\mu_n \equiv EX^n, \tag{2.7}$$

provided the expectation is well defined.

Definition 6.2.10: The variance of a random variable X is defined as $\mathrm{Var}(X) = E(X - EX)^2$, provided $EX^2 < \infty$.

Definition 6.2.11: The *moment generating function (mgf)* of a random variable X is defined by

$$M_X(t) \equiv E(e^{tX}) \quad \text{for all} \quad t \in \mathbb{R}. \tag{2.8}$$

Since e^{tX} is always nonnegative, $E(e^{tX})$ is well defined but could be infinity. Proposition 6.2.1 gives a way of computing the moments and the mgf of X without explicitly computing the distribution of X^k or e^{tX}. As an illustration, consider the case of a random variable X defined on the probability space $(\Omega, \mathcal{F}, P)$ with $\Omega = \{H, T\}^n$, $n \in \mathbb{N}$, $\mathcal{F} =$ the power set of Ω and $P =$ the probability distribution defined by

$$P(\{\omega\}) = p^{X(\omega)} q^{n - X(\omega)}$$

where $0 < p < 1$, $q = 1 - p$, and $X(\omega) =$ the number of H's in ω. By the change of variable formula, the mgf of X is given by

$$\begin{aligned} M_X(t) &\equiv \int e^{tx} P_X(dx) \\ &= \sum_{r=0}^{n} e^{tr} \binom{n}{r} p^r q^{n-r} = (pe^t + q)^n, \end{aligned}$$

since P_X, the probability distribution of X, is supported on $\{0, 1, 2, \ldots, n\}$ with $P_X(\{r\}) = \binom{n}{r} p^r q^{n-r}$. Note that P_X is the Binomial (n, p) distribution. Here, $M_X(t)$ is computed using the distribution of X, i.e., using the middle term in (2.6) only.

The connection between the mgf $M_X(\cdot)$ and the moments of a random variable X is given in the following propositions.

Proposition 6.2.2: *Let X be a nonnegative random variable and $t \geq 0$. Then*

$$M_X(t) \equiv E(e^{tX}) = \sum_{n=0}^{\infty} \frac{t^n \mu_n}{n!} \tag{2.9}$$

where μ_n is as in (2.7).

Proof: Since $e^{tX} = \sum_{n=0}^{\infty} t^n \frac{X^n}{n!}$ and X is nonnegative, (2.9) follows from the MCT. $\square$

Proposition 6.2.3: *Let X be a random variable and let $M_X(t)$ be finite for all $|t| < \epsilon$, for some $\epsilon > 0$. Then*

(i) $E|X|^n < \infty$ *for all* $n \geq 1$,

(ii) $M_X(t) = \sum_{n=0}^{\infty} t^n \frac{\mu_n}{n!}$ *for all* $|t| < \epsilon$,

(iii) $M_X(\cdot)$ *is infinitely differentiable on* $(-\epsilon, +\epsilon)$ *and for* $r \in \mathbb{N}$, *the* r*th derivative of* $M_X(\cdot)$ *is*

$$M_X^{(r)}(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mu_{n+r} = E(e^{tX} X^r) \quad \text{for} \quad |t| < \epsilon. \tag{2.10}$$

In particular,

$$M_X^{(r)}(0) = \mu_r = EX^r. \tag{2.11}$$

Proof: Since $M_X(t) < \infty$ for all $|t| < \epsilon$,

$$E(e^{|tX|}) \leq E(e^{tX}) + E(e^{-tX}) < \infty \quad \text{for} \quad |t| < \epsilon. \tag{2.12}$$

Also, $e^{|tX|} \geq \frac{|t|^n |X|^n}{n!}$ for all $n \in \mathbb{N}$ and hence, (i) follows by choosing a t in $(0, \epsilon)$. Next note that $\left|\sum_{j=0}^{n} \frac{(tx)^j}{j!}\right| \leq e^{|tx|}$ for all x in $\mathbb{R}$ and all $n \in \mathbb{N}$. Hence, by (2.12) and the DCT, (ii) follows.

Turning to (iii), since $M_X(\cdot)$ admits a power series expansion convergent in $|t| < \epsilon$, it is infinitely differentiable in $|t| < \epsilon$ and the derivatives of $M_X(\cdot)$ can be found by term-by-term differentiation of the power series (see Rudin (1976), Chapter 9). Hence,

$$\begin{aligned} M_X^{(r)}(t) &= \frac{d^r}{dt^r} \left(\sum_{n=0}^{\infty} \frac{t^n \mu_n}{n!} \right) \\ &= \sum_{n=0}^{\infty} \frac{\mu_n}{n!} \frac{d^r(t^n)}{dt^r} \\ &= \sum_{n=r}^{\infty} \mu_n \frac{t^{n-r}}{(n-r)!} \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \mu_{n+r} . \end{aligned}$$

The verification of the second equality in (2.10) is left in an exercise (see Problem 6.4). $\square$

Remark 6.2.3: If the mgf $M_X(\cdot)$ is finite for $|t| < \epsilon$ for some $\epsilon > 0$, then by part (ii) of the above proposition, $M_X(t)$ has a power series expansion in t around 0 and $\frac{\mu_n}{n!}$ is simply the coefficient of t^n. For example, if X has a $N(0,1)$ distribution, then for all $t \in \mathbb{R}$,

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = e^{t^2/2} \\ &= \sum_{k=0}^{\infty} \frac{(t^2)^k}{k!} \frac{1}{2^k} \ . \end{aligned} \tag{2.13}$$

Thus, $\mu_n = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \frac{(2k)!}{k!2^k} & \text{if } n = 2k, k = 1, 2, \ldots \end{cases}$

Remark 6.2.4: If $M_X(t)$ is finite for $|t| < \epsilon$ for some $\epsilon > 0$, then all the moments $\{\mu_n\}_{n \geq 1}$ of X are determined and also its probability distribution. However, in general, the sequence $\{\mu_n\}_{n \geq 1}$ of moments of X need not determine the distribution of X uniquely.

Table 6.2.1 gives the mean, variance, and the mgf of a number of standard probability distributions on the real line.

For future reference, some of the inequalities established in Section 3.1 are specialized for random variables and collected below without proofs.

Proposition 6.2.4: (*Markov's inequality*). *Let X be a random variable on $(\Omega, \mathcal{F}, P)$. Then for any $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ nondecreasing and any $t > 0$ with $\phi(t) > 0$,*

$$P(|X| \geq t) \leq \frac{E\big(\phi(|X|)\big)}{\phi(t)} \ . \tag{2.14}$$

In particular,

(i) for $r > 0$, $t > 0$,

$$P(X \geq t) \leq P(|X| \geq t) \leq \frac{E|X|^r}{t^r}, \tag{2.15}$$

(ii) for any $t \geq 0$,

$$P(|X| \geq t) \leq \frac{E(e^{\theta|X|})}{e^{\theta t}} \ ,$$

for any $\theta > 0$ and hence

$$P(|X| \geq t) \leq \inf_{\theta > 0} \frac{E(e^{\theta|X|})}{e^{\theta t}} \ . \tag{2.16}$$

Proposition 6.2.5: (*Chebychev's inequality*). *Let X be a random variable with $EX^2 < \infty$, $EX = \mu$, $Var(X) = \sigma^2$. Then for any $k > 0$,*

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \ . \tag{2.17}$$

TABLE 6.2.1. Mean, variance, mgf of the distributions listed in Tables 4.6.1 and 4.6.2.

Distribution	Mean	Variance	mgf M(t)
Bernoulli (p), $0 < p < 1$	p	$p(1-p)$	$(1-p)+pe^t$, $t \in \mathbb{R}$
Binomial (n,p), $p \in (0,1),\ n \in \mathbb{N}$	np	$np(1-p)$	$\big((1-p)+pe^t\big)^n$, $t \in \mathbb{R}$
Geometric (p), $p \in (0,1)$	$\frac{1}{p}$	$\frac{1-p}{p}$	$\frac{pe^t}{1-(1-p)e^t}$, $t \in \big(-\infty, -\log(1-p)\big)$
Poisson (λ) $\lambda \in (0,\infty)$	λ	λ	$\exp\big(\lambda(e^t-1)\big)$, $t \in \mathbb{R}$
Uniform (a,b), $-\infty < a < b < \infty$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{(b-a)t}$, $t \in \mathbb{R}\setminus\{0\}$; $M(0)=1$
Exponential (β), $\beta \in (0,\infty)$	β	β^2	$\frac{1}{1-\beta t}$, $t \in (-\infty, \frac{1}{\beta})$
Gamma (α,β), $\alpha,\beta \in (0,\infty)$	$\alpha\beta$	$\alpha\beta^2$	$(1-\beta t)^{-\alpha}$, $t \in (-\infty, \frac{1}{\beta})$
Beta (α,β), $\alpha,\beta \in (0,\infty)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\Big[1+\sum_{k=1}^{\infty}\Big(\prod_{r=0}^{k-1}\frac{\alpha+r}{\alpha+\beta+r}\Big)\frac{t^k}{k!}\Big]$, $t \in \mathbb{R}$
Cauchy (γ,σ), $\gamma \in \mathbb{R},\ \sigma \in (0,\infty)$	not defined since $E\|X\| = \infty$	not defined since $E\|X\|^2 = \infty$	∞ for all $t \neq 0$
Normal (γ,σ^2), $\gamma \in \mathbb{R},\ \sigma \in (0,\infty)$	γ	σ^2	$\exp(\gamma t + \frac{t^2\sigma^2}{2})$, $t \in \mathbb{R}$
Lognormal (γ,σ^2), $\gamma \in \mathbb{R},\ \sigma \in (0,\infty)$	$e^{\gamma+\frac{\sigma^2}{2}}$	$[e^{2(\gamma+\sigma^2)} - e^{2\gamma+\sigma^2}]$	∞ for all $t > 0$

Proposition 6.2.6: (*Jensen's inequality*). *Let X be a random variable with $P(a < X < b) = 1$ for $-\infty \le a < b \le \infty$. Let $\phi : (a,b) \to \mathbb{R}$ be convex on (a,b). Then*

$$E\phi(X) \ge \phi(EX), \tag{2.18}$$

provided $E|X| < \infty$ and $E|\phi(X)| < \infty$.

Proposition 6.2.7: (*Hölder's inequality*). *Let X and Y be random variables on $(\Omega, \mathcal{F}, P)$ with $E|X|^p < \infty$, $E|Y|^q < \infty$, $1 < p < \infty$, $1 < q < \infty$, $\frac{1}{p} + \frac{1}{q} = 1$. Then*

$$E|XY| \leq (E|X|^p)^{1/p}(E|Y|^q)^{1/q}, \tag{2.19}$$

with equality holding iff $P(c_1|X|^p = c_2|Y|^q) = 1$ for some $0 \leq c_1, c_2 < \infty$.

Proposition 6.2.8: (*Cauchy-Schwarz inequality*). *Let X and Y be random variables on $(\Omega, \mathcal{F}, P)$ with $E|X|^2 < \infty, E|Y|^2 < \infty$. Then*

$$|Cov(X, Y)| \leq \sqrt{Var(X)}\sqrt{Var(Y)}\ , \tag{2.20}$$

where $Cov(X, Y) = EXY - EXEY$. If $Var(X) > 0$, then equality holds in (2.20) iff $P(Y = aX + b) = 1$ for some constants a, b in $\mathbb{R}$ (Problem 6.6).

Proposition 6.2.9: (*Minkowski's inequality*). *Let X and Y be random variables on $(\Omega, \mathcal{F}, P)$ such that $E|X|^p < \infty, E|Y|^p < \infty$ for some $1 \leq p < \infty$. Then*

$$(E|X + Y|^p)^{1/p} \leq (E|X|^p)^{1/p} + (E|Y|^p)^{1/p}. \tag{2.21}$$

Definition 6.2.12: (*Product moments of random vectors*). Let $X = (X_1, X_2, \ldots, X_k)$ be a random vector. The *product moment of order* $r = (r_1, r_2, \ldots, r_k)$, with r_i being a nonnegative integer for each i, is defined as

$$\mu_r \equiv \mu_{r_1, r_2, \ldots, r_k} \equiv E(X_1^{r_1} X_2^{r_2} \cdots X_k^{r_k}), \tag{2.22}$$

provided $E|X_1^{r_1} \cdots X_k^{r_k}| < \infty$. The *joint moment generating function (joint mgf)* of a random vector $X = (X_1, X_2, \ldots, X_k)$ is defined by

$$M_{X_1, \ldots, X_k}(t_1, t_2, \ldots, t_k) \equiv E(e^{t_1 X_1 + t_2 X_2 + \cdots + t_k X_k}), \tag{2.23}$$

for all $t_1, t_2, \ldots, t_k$ in $\mathbb{R}$.

As in the case of a random variable, if the joint mgf $M_{X_1, X_2, \ldots, X_k}(t_1, \ldots, t_k)$ is finite for all $(t_1, t_2, \ldots, t_k)$ with $|t_i| < \epsilon$ for all $i = 1, 2, \ldots, k$ for some $\epsilon > 0$, then an analog of Proposition 6.2.3 holds. For example, the following assertions are valid (cf. Problem 6.4):

(i)
$$E|X_i|^n < \infty \quad \text{for all} \quad i = 1, 2, \ldots, k \quad \text{and} \quad n \geq 1. \tag{2.24}$$

(ii) For $t = (t_1, \ldots, t_k) \in \mathbb{R}^k$ and $r = (r_1, r_2, \ldots, r_k) \in \mathbb{Z}_+^k$, let

$$\begin{aligned} t^r &= t_1^{r_1} t_2^{r_2} \cdots t_k^{r_k}, \\ r! &= r_1! r_2! \cdots r_k!, \quad \text{and} \\ \mu_r &= EX_1^{r_1} X_2^{r_2} \cdots X_k^{r_k}. \end{aligned}$$

Then,

$$M_X(t_1, \ldots, t_k) = \sum_{r \in \mathbb{Z}_+^k} \frac{t^r}{r!} \mu_r \tag{2.25}$$

for all $t = (t_1, t_2, \ldots, t_k) \in (-\epsilon, +\epsilon)^k$.

(iii) For any $r = (r_1, \ldots, r_k) \in \mathbb{Z}_+^k$,

$$\frac{d^r}{dt^r} M_X(t)\Big|_{t=0} = \mu_r, \tag{2.26}$$

where $\frac{d^r}{dt^r} = \frac{\partial^{r_1}}{\partial t_1^{r_1}} \frac{\partial^{r_2}}{\partial t_2^{r_2}} \cdots \frac{\partial^{r_k}}{\partial t_k^{r_k}}$.

6.3 Kolmogorov's consistency theorem

In the previous section, the case of a single random variable and that of a finite dimensional random vector were discussed. The goal of this section is to discuss infinite families of random variables such as a random sequence $\{X_n\}_{n \geq 1}$ or a random function $\{X(t) : 0 \leq t < T\}, 0 \leq T \leq \infty$. For example, X_n could be the population size of the nth generation of a randomly evolving biological population, and $X(t)$ could be the temperature at time t in a chemical reaction over a period $[0, T]$. An example from modeling of spatial random phenomenon is a collection $\{X(s) : s \in S\}$ of random variables $X(s)$ where S is a specified region such as the U.S., and $X(s)$ is the amount of rainfall at location $s \in S$ during a specified month.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\{X_\alpha : \alpha \in A\}$ be a collection of random variables defined on $(\Omega, \mathcal{F}, P)$, where A is a nonempty set. Then for any $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k, 1 \leq k < \infty$, the random vector $(X_{\alpha_1}, X_{\alpha_2}, \ldots, X_{\alpha_k})$ has a joint probability distribution $\mu_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}$ over $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$.

Definition 6.3.1: A (real valued) *stochastic process* with index set A is a family $\{X_\alpha : \alpha \in A\}$ of random variables defined on a probability space $(\Omega, \mathcal{F}, P)$.

Example 6.3.1: (*Examples of stochastic processes*). Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, $P =$ the Lebesgue measure on $[0, 1]$. Let $A_1 = \{1, 2, 3, \ldots\}$, $A_2 = [0, T]$, $0 < T < \infty$. For $\omega \in \Omega$, $n \in A_1$, $t \in A_2$, let

$$\begin{aligned}
X_n(\omega) &= \sin 2\pi n\omega \\
Y_t(\omega) &= \sin 2\pi t\omega \\
Z_n(\omega) &= n\text{th digit in the decimal expansion of } \omega \\
V_{n,t}(\omega) &= X_n^2(\omega) + Y_t^2(\omega).
\end{aligned}$$

Then $\{X_n : n \in A_1\}$, $\{Z_n : n \in A_1\}$, $\{V_{n,t} : (n, t) \in A_1 \times A_2\}$, $\{Y_t : t \in A_2\}$ are all stochastic processes.

Note that a real valued stochastic process $\{X_\alpha : \alpha \in A\}$ may also be viewed as a random real valued function on the set A by the identification $\omega \to f(\omega, \cdot)$, where $f(\omega, \alpha) = X_\alpha(\omega)$ for α in A.

Definition 6.3.2: The family $\{\mu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)}(\cdot) \equiv P((X_{\alpha_1}, \ldots, X_{\alpha_k}) \in \cdot)$: $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k, 1 \leq k < \infty\}$ of probability distributions is called the *family of finite dimensional distributions* (*fdds*) associated with the stochastic process $\{X_\alpha : \alpha \in A\}$.

This family of finite dimensional distributions satisfies the following *consistency* conditions: For any $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k$, $2 \leq k < \infty$, and any $B_1, B_2, \ldots, B_k$ in $\mathcal{B}(\mathbb{R})$,

C1: $\mu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)}(B_1 \times \cdots \times B_{k-1} \times \mathbb{R}) = \mu_{(\alpha_1,\alpha_2,\ldots,\alpha_{k-1})}(B_1 \times \cdots \times B_{k-1})$;

C2: For any permutation $(i_1, i_2, \ldots, i_k)$ of $(1, 2, \ldots, k)$,

$$\mu_{(\alpha_{i_1},\alpha_{i_2},\ldots,\alpha_{i_k})}(B_{i_1} \times B_{i_2} \times \cdots \times B_{i_k}) = \mu_{(\alpha_1,\ldots,\alpha_k)}(B_1 \times B_2 \times \cdots \times B_k)\ .$$

To verify C1, note that

$$\begin{aligned}
&\mu_{(\alpha_1,\alpha_2,\ldots,,\alpha_k)}(B_1 \times B_2 \times \cdots \times B_{k-1} \times \mathbb{R}) \\
&\quad= P(X_{\alpha_1} \in B_1, X_{\alpha_2} \in B_2, \ldots, X_{\alpha_{k-1}} \in B_{k-1}, X_{\alpha_k} \in \mathbb{R}) \\
&\quad= P(X_{\alpha_1} \in B_1, X_{\alpha_2} \in B_2, \ldots, X_{\alpha_{k-1}} \in B_{k-1}) \\
&\quad= \mu_{(\alpha_1,\alpha_2,\ldots,\alpha_{k-1})}(B_1 \times B_2 \times \cdots \times B_{k-1}).
\end{aligned}$$

Similarly, to verify C2, note that

$$\begin{aligned}
&\mu_{(\alpha_{i_1},\alpha_{i_2},\ldots,\alpha_{i_k})}(B_{i_1} \times B_{i_2} \cdots \times B_{i_k}) \\
&\quad= P(X_{\alpha_{i_1}} \in B_{i_1}, X_{\alpha_{i_2}} \in B_{i_2}, \ldots, X_{\alpha_{i_k}} \in B_{i_k}) \\
&\quad= P(X_{\alpha_1} \in B_1, X_{\alpha_2} \in B_2, \ldots, X_{\alpha_k} \in B_k) \\
&\quad= \mu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)}(B_1 \times B_2 \times \cdots \times B_k).
\end{aligned}$$

A natural question is that given a family of probability distributions $Q_A \equiv \{\nu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)} : (\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k,\ 1 \leq k < \infty\}$ on finite dimensional Euclidean spaces, does there exist a real valued stochastic process $\{X_\alpha : \alpha \in A\}$ such that its family of finite dimensional distributions coincides with Q_A?

Kolmogorov (1956) showed that if Q_A satisfies C1 and C2, then such a stochastic process does exist. This is known as Kolmogorov's consistency theorem (also known as Kolmogorov's existence theorem).

Theorem 6.3.1: (*Kolmogorov's consistency theorem*). *Let A be a nonempty set. Let $Q_A \equiv \{\nu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)} : (\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k, 1 \leq k < \infty\}$ be a family of probability distributions such that for each $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k$, $1 \leq k < \infty$,* *(i)* $\nu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)}$ *is a probability distribution on* $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$,

(ii) C1 and C2 hold, i.e., for all $B_1, B_2, \ldots, B_k \in \mathcal{B}(\mathbb{R})$, $2 \leq k < \infty$,

$$\begin{aligned} &\nu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)}(B_1 \times B_2 \times \cdots \times B_{k-1} \times \mathbb{R}) \\ &\quad = \nu_{(\alpha_1,\alpha_2,\ldots,\alpha_{k-1})}(B_1 \times B_2 \times \cdots \times B_{k-1}) \end{aligned} \tag{3.1}$$

and for any permutation $(i_1, i_2, \ldots, i_k)$ *of* $(1, 2, \ldots, k)$,

$$\begin{aligned} &\mu_{(\alpha_{i_1},\alpha_{i_2},\ldots,\alpha_{i_k})}(B_{i_1} \times B_{i_2} \times \cdots \times B_{i_k}) \\ &\quad = \mu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)}(B_1 \times B_2 \times \cdots \times B_k). \end{aligned} \tag{3.2}$$

Then, there exists a probability space $(\Omega, \mathcal{F}, P)$ *and a stochastic process* $X_A \equiv \{X_\alpha : \alpha \in A\}$ *on* $(\Omega, \mathcal{F}, P)$ *such that* Q_A *is the family of finite dimensional distributions associated with* X_A.

Remark 6.3.1: Thus the above theorem says that given the family Q_A satisfying conditions (i) and (ii), there exists a real valued function on $A \times \Omega$ such that for each ω, $f(\cdot, \omega)$ is a function on A and for each $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k$, the vector $\big(f(\alpha_1, \omega), f(\alpha_2, \omega), \ldots, f(\alpha_k, \omega)\big)$ is a random vector with probability distribution $\nu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)}$. This random function point of view is useful in dealing with functionals of the form $M(\omega) \equiv \{\sup f(\alpha, \omega) : \alpha \in A\}$. For example, if $A_1 = \{1, 2, \ldots\}$, then one might consider functionals such as $\lim_{n\to\infty} f(n, \omega)$, $\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} f(j, \omega)$, $\sum_{j=1}^{\infty} f(j, \omega)$, etc. Since the random functionals are not fully determined by $f(\alpha, \omega)$ for finitely many α's, it is not possible to compute probabilities of events defined in terms of these functionals from the knowledge of the finite dimensional distribution of $(f(\alpha_1, \omega), \ldots, f(\alpha_k, \omega))$ for a given $(\alpha_1, \ldots, \alpha_k)$, no matter how large k is. Kolmogorov's consistency theorem allows one to compute these probabilities given *all* finite dimensional distributions (provided that the functionals satisfy appropriate measurability conditions).

Given a probability measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, now consider the problem of constructing a probability space $(\Omega, \mathcal{F}, P)$ and a random variable X on it with distribution μ. A natural solution is to set the sample space Ω to be $\mathbb{R}$, the σ-algebra $\mathcal{F}$ to be $\mathcal{B}(\mathbb{R})$, and the probability measure P to be μ and the random variable X to be the identity map $X(\omega) \equiv \omega$. Similarly, given a probability measure μ on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R})^k)$, one can set the sample space Ω to be $\mathbb{R}^k$ and the σ-algebra $\mathcal{F}$ to be $\mathcal{B}(\mathbb{R}^k)$ and the probability measure P to be μ and the random vector X to be the identity map.

Arguing in the same fashion, given a family Q_A of finite dimensional distributions with index set A, to construct a stochastic process $\{X_\alpha : \alpha \in A\}$ with index set A on some probability space $(\Omega, \mathcal{F}, P)$, it is natural to set the sample space Ω to be $\mathbb{R}^A$, the collection of all real valued functions on A, $\mathcal{F}$ to be a suitable σ-algebra that includes all finite dimensional events,

P to be an appropriate probability measure that yields Q_A, and X to be the identity map.

These considerations lead to the following definitions.

Definition 6.3.3: Let A be a nonempty set. Then $\mathbb{R}^A \equiv \{f \mid f : A \to \mathbb{R}\}$, the collection of all real valued functions on A.

If A is a finite set $\{a_1, a_2, \ldots, a_k\}$, then $\mathbb{R}^A$ can be identified with $\mathbb{R}^k$ by associating each $f \in \mathbb{R}^A$ with the vector $(f(a_1), f(a_2), \ldots, f(a_k))$ in $\mathbb{R}^k$. If A is a countably infinite set $\{a_1, a_2, a_3, \ldots\}$, then $\mathbb{R}^A$ can be similarly identified with $\mathbb{R}^\infty$, the set of all sequences $\{x_1, x_2, x_3, \ldots\}$ of real numbers. If A is the interval $[0, 1]$, then $\mathbb{R}^A$ is the collection of all real valued functions on $[0, 1]$.

Definition 6.3.4: Let A be a nonempty set. A subset $C \subset \mathbb{R}^A$ is called a *finite dimensional cylinder set* (*fdcs*) if there exists a finite subset $A_1 \subset A$, say, $A_1 \equiv \{\alpha_1, \alpha_2, \ldots, \alpha_k\}, 1 \leq k < \infty$ and a Borel set B in $\mathcal{B}(\mathbb{R}^k)$ such that $C = \{f : f \in \mathbb{R}^A \text{ and } (f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_k)) \in B\}$. The set B is called a *base* for C.

The collection of all finite dimensional cylinder sets will be denoted by $\mathcal{C}$.

The name *cylinder* is motivated by the following example:

Example 6.3.2: Let $A = \{1, 2, 3\}$ and $C = \{(x_1, x_2, x_3) : x_1^2 + x_2^2 \leq 1\}$. Then C is a cylinder (in the usual sense of the English word), but with infinite height and depth. According to Definition 6.3.4, C is also a cylinder in $\mathbb{R}^3$ with the unit circle in $\mathbb{R}^2$ as its base.

Examples 6.3.3 and 6.3.4 below are examples of fdcs, whereas Example 6.3.5 is an example of a set that is not a fdcs.

Example 6.3.3: Let $A = \{1, 2\}$ and $C = \{(x_1, x_2) : |\sin 2\pi x_1| \leq \frac{1}{\sqrt{2}}\}$.

Example 6.3.4: Let $A = \{1, 2, 3, \ldots\}$ and $C = \{(x_1, x_2, x_3, \ldots) : \frac{x_{17}^2}{4} + \frac{x_{30}^2}{10} - \frac{x_{42}^2}{5} \leq 10\}$.

Example 6.3.5: Let $A = \{1, 2, 3, \ldots\}$ and $D = \{(x_1, x_2, x_3, \ldots) : x_j \in \mathbb{R}$ for all $j \geq 1$ and $\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^n x_j$ exists$\}$ is *not* a finite dimensional cylinder set (Problem 6.8).

Proposition 6.3.2: *Let A be a nonempty set and $\mathcal{C}$ be the collection of all finite dimensional cylinder sets in $\mathbb{R}^A$. Then $\mathcal{C}$ is an algebra.*

Proof: Let $C_1, C_2 \in \mathcal{C}$ and let

$$\begin{aligned}
C_1 &= \{f : f \in \mathbb{R}^A \text{ and } \big(f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_k)\big) \in B_1\} \\
C_2 &= \{f : f \in \mathbb{R}^A \text{ and } \big(f(\beta_1), f(\beta_2), \ldots, f(\beta_j)\big) \in B_2\}
\end{aligned}$$

for some $A_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_k\} \subset A, A_2 = \{\beta_1, \beta_2, \ldots, \beta_j\} \subset A, B_1 \in \mathcal{B}(\mathbb{R}^k), B_2 \in \mathcal{B}(\mathbb{R}^j), 1 \leq k < \infty, 1 \leq j < \infty$. Let $A_3 = A_1 \cup A_2 = \{\gamma_1, \gamma_2, \ldots, \gamma_\ell\}$, where without loss of generality $(\gamma_1, \gamma_2, \ldots, \gamma_k) = (\alpha_1, \alpha_2, \ldots, \alpha_k)$ and $(\gamma_{\ell-j+1}, \ldots, \gamma_{\ell-1}, \gamma_\ell) = (\beta_1, \beta_2, \ldots, \beta_j)$. Then C_1 and C_2 may be expressed as

$$\begin{aligned} C_1 &= \{f : f \in \mathbb{R}^A \text{ and } \big(f(\gamma_1), f(\gamma_2), \ldots, f(\gamma_\ell)\big) \in \tilde{B}_1\} \\ C_2 &= \{f : f \in \mathbb{R}^A \text{ and } \big(f(\gamma_1), \ldots, f(\gamma_\ell)\big) \in \tilde{B}_2\} \end{aligned}$$

where $\tilde{B}_1 = B_1 \times \mathbb{R}^{\ell-k}$ and $\tilde{B}_2 = \mathbb{R}^{\ell-j} \times B_2$. Thus, $C_1 \cup C_2 = \{f : f \in \mathbb{R}^A$ and $(f(\gamma_1), \ldots, f(\gamma_\ell)) \in \tilde{B}_1 \cup \tilde{B}_2\}$. Since both $\tilde{B}_1$ and $\tilde{B}_2$ lie in $\mathcal{B}(\mathbb{R}^\ell)$, $C_1 \cup C_2 \in \mathcal{C}$.

Next note that, $C_1^c = \{f : f \in \mathbb{R}^A$ and $\big(f(\alpha_1), \ldots, f(\alpha_k)\big) \in B_1^c\}$. Since $B_1^c \in \mathcal{B}(\mathbb{R}^k)$, it follows that $C_1^c \in \mathcal{C}$. Thus, $\mathcal{C}$ is an algebra. $\square$

Remark 6.3.2: If A is a *finite* nonempty set, the collection $\mathcal{C}$ is also a σ-algebra.

Definition 6.3.5: Let A be a nonempty set. Let $\mathcal{R}^A$ be the σ-algebra generated by the collection $\mathcal{C}$. Then $\mathcal{R}^A$ is called the *product σ-algebra* on $\mathbb{R}^A$.

Remark 6.3.3: If $A = \{1, 2, 3, \ldots\} \equiv \mathbb{N}$ and $\mathbb{R}^{\mathbb{N}}$ is identified with the set $\mathbb{R}^\infty$ of all sequences of real numbers, then the product σ-algebra $\mathcal{R}^{\mathbb{N}}$ coincides with the Borel σ-algebra $\mathcal{B}(\mathbb{R}^\infty)$ on $\mathbb{R}^\infty$ under the metric

$$d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} \left(\frac{|x_j - y_j|}{1 + |x_j - y_j|} \right) \tag{3.3}$$

for $x = (x_1, x_2, \ldots)$, $y = (y_1, y_2, \ldots)$ in $\mathbb{R}^\infty$ (Problem 6.9).

Definition 6.3.6: Let A be a nonempty set. For any $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k$, $1 \leq k < \infty$, the projection map $\pi_{(\alpha_1, \ldots, \alpha_k)}$ from $\mathbb{R}^A$ to $\mathbb{R}^k$ is defined by

$$\pi_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}(f) = (f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_k)). \tag{3.4}$$

In particular, for $\alpha \in A$,

$$\pi_\alpha(f) = f(\alpha) \tag{3.5}$$

is called a *co-ordinate* map.

The projection map π_{A_1} for any arbitrary subset $A_1 \subset A$ may be similarly defined. The next proposition follows from the definition of $\mathcal{R}^A$.

Proposition 6.3.3:

(i) *For each $\alpha \in A$, the map π_α from $\mathbb{R}^A$ to $\mathbb{R}$ is $\langle \mathcal{R}^A, \mathcal{B}(\mathbb{R}) \rangle$-measurable.*

(ii) *For any $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in A^k$, $1 \leq k < \infty$, the map $\pi_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}$ from $\mathbb{R}^A$ to $\mathbb{R}^k$ is $\langle \mathcal{R}^A, \mathcal{B}(\mathbb{R}^k) \rangle$-measurable.*

Proof of Theorem 6.3.1: Let $\Omega = \mathbb{R}^A$ and $\mathcal{F} \equiv \mathcal{R}^A$. Define a set function P on $\mathcal{C}$ by

$$P(C) = \mu_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}(B) \tag{3.6}$$

for a C in $\mathcal{C}$ with representation

$$C = \{\omega : \omega \in \mathbb{R}^A,\ \big(\omega(\alpha_1), \omega(\alpha_2), \ldots, \omega(\alpha_k)\big) \in B\}. \tag{3.7}$$

The main steps in the proof are

(i) To show that $P(C)$ as defined in (3.6) is independent of the representation (3.7) of C, and

(ii) $P(\cdot)$ is countably additive on $\mathcal{C}$.

Next, by the Caratheodory extension theorem (Theorem 1.3.3), there exists a unique extension of P (also denoted by P) to $\mathcal{F}$ such that $(\Omega, \mathcal{F}, P)$ is a probability space. Defining $X_\alpha(\omega) \equiv \pi_\alpha(\omega) = \omega(\alpha)$ for α in A yields a stochastic process $\{X_\alpha : \alpha \in A\}$ on the probability space

$$(\mathbb{R}^A, \mathcal{R}^A, P) \equiv (\Omega, \mathcal{F}, P)$$

with the family Q_A as its set of finite dimensional distributions. Hence, it remains to establish (i) and (ii). Let $C \in \mathcal{C}$ admit two representations:

$$\begin{aligned} C &\equiv \{\omega : \big(\omega(\alpha_1), \omega(\alpha_2), \ldots, \omega(\alpha_k)\big) \in B_1\} \\ &\equiv \pi_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}(B_1) \end{aligned}$$

and

$$\begin{aligned} C &\equiv \{\omega : \big(\omega(\beta_1), \omega(\beta_2), \ldots, \omega(\beta_j)\big) \in B_2\} \\ &\equiv \pi^{-1}_{(\beta_1, \beta_2, \ldots, \beta_j)}(B_2) \end{aligned}$$

for some $A_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_k\} \subset A$, $1 \leq k < \infty$, and some $A_2 = \{\beta_1, \beta_2, \ldots, \beta_j\} \subset A$, $1 \leq j < \infty$, $B_1 \in \mathcal{B}(\mathbb{R}^k)$ and $B_2 \in \mathcal{B}(\mathbb{R}^j)$. Let $A_3 = A_1 \cup A_2 = \{\gamma_1, \gamma_2, \ldots, \gamma_\ell\}$ and w.l.o.g., let $(\gamma_1, \gamma_2, \ldots, \gamma_k) = (\alpha_1, \alpha_2, \ldots, \alpha_k)$ and $(\gamma_{\ell-j+1}, \gamma_{\ell-j+2}, \gamma_{\ell-1}, \gamma_\ell) = (\beta_1, \beta_2, \ldots, \beta_j)$. Then C may be represented as

$$\begin{aligned} C &= \pi^{-1}_{\gamma_1, \gamma_2, \ldots, \gamma_\ell}(\tilde{B}_1) \\ &= \pi^{-1}_{\gamma_1, \gamma_2, \ldots, \gamma_\ell}(\tilde{B}_2) \end{aligned}$$

where $\tilde{B}_1 = B_1 \times \mathbb{R}^{\ell-k}$ and $\tilde{B}_2 = \mathbb{R}^{\ell-j} \times B_2$. Note that $(\omega(\gamma_1), \ldots, \omega(\gamma_\ell)) \in \tilde{B}_1$ iff $\omega \in C$ iff $(\omega(\gamma_1), \ldots, \omega(\gamma_\ell)) \in \tilde{B}_2$ and thus

$$\tilde{B}_1 = \tilde{B}_2. \tag{3.8}$$

Next by the first consistency condition (3.1) and induction,

$$\nu_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\tilde{B}_1) = \nu_{(\alpha_1,\alpha_2,\ldots,\alpha_k)}(B_1). \tag{3.9}$$

Also by (3.2), for B_2 of the form $B_{21} \times B_{22} \times \cdots \times B_{2j}$ with $B_{2i} \in \mathcal{B}(\mathbb{R})$ for all $1 \le i \le j$,

$$\nu_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\mathbb{R}^{\ell-j} \times B_2) = \nu_{(\gamma_{\ell-j+1},\ldots,\gamma_\ell,\gamma_1,\gamma_2,\ldots,\gamma_{\ell-j})}(B_2 \times \mathbb{R}^{\ell-j}).$$

Now note that

(a) $\nu_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\mathbb{R}^{\ell-j} \times B)$ and $\nu_{(\gamma_{\ell-j+1},\ldots,\gamma_\ell,\gamma_1,\gamma_2,\ldots,\gamma_{\ell-j})}(B \times \mathbb{R}^{\ell-j})$, considered as set functions defined for $B \in \mathcal{B}(\mathbb{R}^j)$, are probability measures on $\mathcal{B}(\mathbb{R}^j)$,

(b) they coincide on the class Γ of sets of the form $B = B_{21} \times B_{22} \times \cdots \times B_{2j}$ with $B_{2i} \in \mathcal{B}(\mathbb{R})$ for all i, and

(c) the class Γ is a π-class and it generates $\mathcal{B}(\mathbb{R}^j)$.

Hence, by the uniqueness theorem (Theorem 1.3.6),

$$\nu_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\mathbb{R}^{\ell-j} \times B) = \nu_{(\gamma_{\ell-j+1},\ldots,\gamma_\ell,\gamma_1,\gamma_2,\ldots,\gamma_{\ell-j})}(B \times \mathbb{R}^{\ell-j}) \tag{3.10}$$

for all $B \in \mathcal{B}(\mathbb{R}^j)$.

Again by (3.1) and induction

$$\begin{aligned} &\nu_{(\gamma_{\ell-j+1},\ldots,\gamma_\ell,\gamma_1,\gamma_2,\ldots,\gamma_{\ell-j})}(B_2 \times \mathbb{R}^{\ell-j}) \\ &= \nu_{(\gamma_{\ell-j+1},\ldots,\gamma_\ell)}(B_2) = \nu_{(\beta_1,\beta_2,\ldots,\beta_j)}(B_2). \end{aligned} \tag{3.11}$$

Since $\tilde{B}_2 = \mathbb{R}^{\ell-j} \times B_2$, by (3.10) and (3.11)

$$\nu_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\tilde{B}_2) = \nu_{(\beta_1,\beta_2,\ldots,\beta_j)}(B_2).$$

Now from (3.8) and (3.9) it follows that

$$\begin{aligned} \nu_{(\alpha_1,\ldots,\alpha_k)}(B_1) &= \nu_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\tilde{B}_1) \\ &= \nu_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\tilde{B}_2) \\ &= \nu_{(\beta_1,\beta_2,\ldots,\beta_j)}(B_2), \end{aligned}$$

thus establishing (i).

To establish (ii), it needs to be shown that

(ii)^a $P(C_1 \cup C_2) = P(C_1) + P(C_2)$ if $C_1, C_2 \in \mathcal{C}$ and $C_1 \cap C_2 = \emptyset$.

(ii)^b $C_n \in \mathcal{C}, C_n \supset C_{n+1}$ for all n, $\bigcap_{n\ge1} C_n = \emptyset \Rightarrow P(C_n) \downarrow 0$.

Let $C_1 = \pi^{-1}_{(\alpha_1,\ldots,\alpha_k)}(B_1)$ and $C_2 = \pi^{-1}_{(\beta_1,\ldots,\beta_j)}(B_2)$ for $B_1 \in \mathcal{B}(\mathbb{R}^k)$, $B_2 \in \mathcal{B}(\mathbb{R}^j)$, $\{\alpha_1,\ldots,\alpha_k\} \subset A$ and $\{\beta_1,\ldots,\beta_j\} \subset A$, $1 \le j,\, k < \infty$.

As in the proof of Proposition 6.3.2, C_1 and C_2 may be represented as

$$C_i = \pi^{-1}_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\tilde{B}_i),\ i = 1,2,$$

where $\tilde{B}_i \in \mathcal{B}(\mathbb{R}^\ell)$. Since C_1 and C_2 are disjoint by hypothesis, it follows that $\tilde{B}_1$ and $\tilde{B}_2$ are disjoint. Also, since

$$P(C_i) = \nu_{(\gamma_1,\gamma_2,\ldots,\gamma_\ell)}(\tilde{B}_i),\ i = 1,2,$$

and $\nu_{(\gamma_1,\ldots,\gamma_\ell)}(\cdot)$ is a measure on $\mathcal{B}(\mathbb{R}^\ell)$, it follows that

$$\begin{aligned}
P(C_1 \cup C_2) &= \nu_{(\gamma_1,\ldots,\gamma_\ell)}(\tilde{B}_1 \cup \tilde{B}_2) \\
&= \nu_{(\gamma_1,\ldots,\gamma_\ell)}(\tilde{B}_1) + \nu_{(\gamma_1,\ldots,\gamma_\ell)}(\tilde{B}_2) \\
&= P(C_1) + P(C_2),
\end{aligned}$$

thus proving (ii)^a.

To prove (ii)^b, note that for any sequence $\{C_n\}_{n\ge1} \subset \mathcal{C}$, there exists a countable set $A_1 = \{\alpha_1,\alpha_2,\ldots,\alpha_n,\ldots\}$, an increasing sequence $\{k_n\}_{n\ge1}$ of positive integers and a sequence of Borel sets $\{B_n\}_{n\ge1}$ such that $B_n \in \mathcal{B}(\mathbb{R}^{k_n})$ and $C_n = \pi^{-1}_{(\alpha_1,\alpha_2,\ldots,\alpha_{k_n})}(B_n)$ for all $n \in \mathbb{N}$. Now suppose that $\{C_n\}_{n\ge1}$ is decreasing. It will be shown that if $\lim_{n\to\infty} P(C_n) = \delta > 0$, then $\bigcap_{n\ge1} C_n \ne \emptyset$. For each n, by the regularity of measures (Corollary 1.3.5), there exists a compact set $G_n \subset B_n$ such that

$$\nu_{(\alpha_1,\ldots,\alpha_{k_n})}(B_n \setminus G_n) < \frac{\delta}{2^{n+1}}\ .$$

Let $D_n = \pi^{-1}_{(\alpha_1,\alpha_2,\ldots,\alpha_{k_n})}(G_n)$. Then $P(C_n \setminus D_n) < \frac{\delta}{2^{n+1}}$. Let $H_n = \bigcap_{j=1}^n D_j$. Then $\{H_n\}_{n\ge1}$ is decreasing and

$$\begin{aligned}
P(C_n \setminus H_n) &= P\big(C_n \cap H_n^c\big) = P\bigg(\bigcup_{j=1}^n (C_n \cap D_j^c)\bigg) \\
&\le \sum_{j=1}^n P(C_n \setminus D_j) \le \sum_{j=1}^n P(C_j \setminus D_j) \\
&\qquad\qquad \text{(since } \{C_n\}_{n\ge1} \text{ is decreasing)} \\
&\le \sum_{j=1}^n \frac{\delta}{2^{j+1}} < \frac{\delta}{2}\ .
\end{aligned}$$

Since $P(C_n) \downarrow \delta > 0$, $H_n \subset C_n$, and $P(C_n \setminus H_n) < \frac{\delta}{2}$, it follows that $P(H_n) > \frac{\delta}{2}$ for all $n \ge 1$. This implies $H_n \ne \emptyset$ for each n. It will now be shown that $\bigcap_{n\ge1} H_n \ne \emptyset$. Let $\{\omega_n\}_{n\ge1}$ be a sequence of elements from $\Omega = \mathbb{R}^A$ such that for each n, $\omega_n \in H_n$. Then, since $\{H_n\}_{n\geq 1}$ is a decreasing sequence, for each $1 \leq j < \infty$, $\omega_n \in H_j$ for $n \geq j$. This implies that the vector $(\omega_n(\alpha_1), \omega_n(\alpha_2), \ldots, \omega_n(\alpha_{k_j})) \in G_j$ for all $n \geq j$. Since G_1 is compact, there exists a subsequence $\{n_{1i}\}_{i\geq 1}$ such that $\lim_{i\to\infty} \omega_{n_{1i}}(\alpha_1) = \omega(\alpha_1)$ exists. Next, since G_2 is compact, there exists a further sequence $\{n_{2i}\}_{i\geq 1}$ of $\{n_{1i}\}_{i\geq 1}$ such that $\lim_{i\to\infty} \omega_{n_{2i}}(\alpha_2) = \omega(\alpha_2)$ exists. Proceeding this way and applying the usual 'diagonal method,' a subsequence $\{n_i\}_{i\geq 1}$ is obtained such that $\lim_{i\to\infty} \omega_{n_i}(\alpha_j) = \omega(\alpha_j)$ for all $1 \leq j < \infty$. Let $\omega(\alpha) = 0$ for $\alpha \notin \{\alpha_1, \alpha_2, \ldots\}$. Since for each j, G_j is compact, $(\omega(\alpha_1), \omega(\alpha_2), \ldots, \omega(\alpha_{k_j})) \in G_j$ and hence $\omega \in H_j$. Thus, $\omega \in \bigcap_{j\geq 1} H_j \subset \bigcap_{j\geq 1} C_j$ implying $\bigcap_{j\geq 1} C_j \neq \emptyset$. The proof of the theorem is now complete. □

When the index set A is countable and identified with the set $\mathbb{N} \equiv \{1, 2, 3, \ldots\}$, it is possible to give a simpler formulation of the consistency conditions.

Theorem 6.3.4: *Let $\{\mu_n\}_{n\geq 1}$ be a sequence of probability measures such that*

(i) for each $n \in \mathbb{N}$, μ_n is a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$,

(ii) for each $n \in \mathbb{N}$, $\mu_{n+1}(B \times \mathbb{R}) = \mu_n(B)$ for all $B \in \mathcal{B}(\mathbb{R}^n)$.

Then there exists a stochastic process $\{X_n : n \geq 1\}$ on a probability space $(\Omega, \mathcal{F}, P)$ with $\Omega = \mathbb{R}^\infty, \mathcal{F} = \mathcal{B}(\mathbb{R}^\infty)$ such that for each $n \geq 1$, the probability distribution $P_{(X_1, X_2, \ldots, X_n)}$ of the random vector $(X_1, X_2, \ldots, X_n)$ is μ_n.

Proof: For any $\{i_1, i_2, \ldots, i_k\} \subset \mathbb{N}$, let $j_1 < j_2 < \cdots < j_k$ be the increasing rearrangement of $i_1, i_2, \ldots, i_k$. Then there exists a permutation $(r_1, r_2, \ldots, r_k)$ of $(1, 2, \ldots, k)$ such that $j_1 = i_{r_1}, j_2 = i_{r_2}, \ldots, j_k = i_{r_k}$.

Now define

$$\nu_{(j_1, j_2, \ldots, j_k)}(\cdot) \equiv \mu_{j_k} \pi^{-1}_{j_1, j_2, \ldots, j_k}(\cdot)$$

where $\pi_{j_1, j_2, \ldots, j_k}(x_1, \ldots, x_{j_k}) = (x_{j_1}, x_{j_2}, \ldots, x_{j_k})$ for all $(x_1, x_2, \ldots, x_{j_k}) \in \mathbb{R}^{j_k}$.

Next define

$$\nu_{(i_1, i_2, \ldots, i_k)}(B_1 \times B_2 \times \ldots \times B_k) \equiv \nu_{(j_1, j_2, \ldots, j_k)}(B_{r_1} \times B_{r_2} \times \ldots \times B_{r_k})$$

where $B_i \in \mathcal{B}(\mathbb{R})$ for all i, $1 \leq i \leq k$. It can be verified that this family of finite dimensional distributions

$$Q_{\mathbb{N}} \equiv \{\nu_{(i_1, i_2, \ldots, i_k)}(\cdot) : \{i_1, i_2, \ldots, i_k\} \subset \mathbb{N},\ 1 \leq k < \infty\} \tag{3.12}$$

satisfies the consistency conditions (3.1) and (3.2) of Theorem 6.3.1 and hence the assertion follows. □

Example 6.3.6: (*Sequence of independent random variables*). Let $\{F_n\}_{n\geq 1}$ be a sequence of cdfs on $\mathbb{R}$. Consider the problem of constructing a sequence $\{X_n\}_{n\geq 1}$ of random variables on a probability space $(\Omega, \mathcal{F}, P)$ such that (i) for each $n \in \mathbb{N}$, X_n has cdf F_n and (ii) for any $n \in \mathbb{N}$ and any $\{i_1, i_2, \ldots, i_n\} \subset \mathbb{N}$, the random variables $\{X_{i_1}, X_{i_2}, \ldots, X_{i_n}\}$ are *independent*, i.e.,

$$P(X_{i_1} \leq x_1, X_{i_2} \leq x_2, \ldots, X_{i_n} \leq x_n) = \prod_{j=1}^{n} F_{i_j}(x_j) \tag{3.13}$$

for all $x_1, x_2, \ldots, x_n$ in $\mathbb{R}$.

This problem can be solved by using Theorem 6.3.4. Let μ_n be the Lebesgue-Stieltjes probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ corresponding to the distribution function

$$F_{1,2,\ldots,n}(x_1, x_2, \ldots, x_n) \equiv \prod_{j=1}^{n} F(x_j), \quad x_1, \ldots, x_n \in \mathbb{R}.$$

It is easy to verify that the family $\{\mu_n : n \geq 1\}$ satisfies (i) and (ii) of Theorem 6.3.4. Hence, there exist a probability measure P on the sequence space $\Omega \equiv \mathbb{R}^\infty$ equipped with σ-algebra $\mathcal{F} \equiv \mathcal{B}(\mathbb{R}^\infty)$ and random variables $X_n(\omega) \equiv \pi_n(\omega) \equiv \omega(n)$, for $\omega = (\omega(1), \omega(2), \ldots)$ in $\mathbb{R}^\infty$, $n \geq 1$, such that (3.13) holds.

Example 6.3.7: (*Family of independent random variables*). Given a family $\{F_\alpha : \alpha \in A\}$ of cdfs on $\mathbb{R}$ for some index set A, a construction similar to Example 6.3.6, but using Theorem 6.3.1 yields the existence of a real valued stochastic process $\{X_\alpha : \alpha \in A\}$ such that for any $\{\alpha_1, \alpha_2, \ldots, \alpha_n\} \subset A$, $1 \leq n < \infty$, the random variables $\{X_{\alpha_1}, X_{\alpha_2}, \ldots, X_{\alpha_n}\}$ are independent, i.e., (3.13) holds.

Example 6.3.8: (*Markov chains*). Let $Q = ((q_{ij}))$ be a $k \times k$ *stochastic matrix* for some $1 < k < \infty$. That is,

(a) for all $1 \leq i$, $j \leq k$, $q_{ij} \geq 0$ and

(b) for each $1 \leq i \leq k$, $\sum_{j=1}^{k} q_{ij} = 1$.

Let $p = (p_1, p_2, \ldots, p_k)$ be a probability vector, i.e., for all i, $p_i \geq 0$, and $\sum_{i=1}^{k} p_i = 1$. Consider the problem of constructing a sequence $\{X_n\}_{n\geq 1}$ of random variables such that for each $n \in \mathbb{N}$,

$$P(X_1 = j_1, X_2 = j_2, \ldots, X_n = j_n) = p_{j_1} q_{j_1 j_2} \ldots q_{j_{n-1} j_n} \tag{3.14}$$

for $1 \leq j_i \leq k$, $i = 1, 2, \ldots, n$.

Let μ_n be the discrete probability distribution determined by the right side of (3.14), that is,

$$\mu_n(\{(j_1, j_2, \ldots, j_n)\}) = p_{j_1} q_{j_1 j_2} \ldots q_{j_{n-1} j_n}$$

for all $(j_1, \ldots, j_n)$ such that $1 \leq j_i \leq k$ for all $1 \leq i \leq n$. It is easy to verify that $\{\mu_n\}_{n \geq 1}$ satisfies the conditions of Theorem 6.3.4 and hence there exist a sequence $\{X_n\}_{n \geq 1}$ of random variables satisfying (3.14). It may be verified that (3.14) is equivalent to

$$\begin{aligned} &P(X_{n+1} = j_{n+1} | X_1 = j_1, \ldots, X_n = j_n) \\ &= q_{j_n j_{n+1}} = P(X_{n+1} = j_{n+1} | X_n = j_n) \end{aligned} \tag{3.15}$$

for all $n \geq 1$, $1 \leq j_i \leq k$, $i = 1, 2, \ldots, n+1$ provided $P(X_1 = j_1, \ldots, X_n = j_n) > 0$ and $P(X_1 = j) = p_j$ for $1 \leq j \leq k$. This says that the conditional distribution of X_{n+1} given $X_1, X_2, \ldots, X_n$ depends only on X_n. This property is known as the *Markov property*, and the sequence $\{X_n\}_{n \geq 1}$ is called a *Markov chain* with state space $\mathbb{S} \equiv \{1, 2, \ldots, k\}$ and *time homogeneous transition probability matrix* $((q_{ij}))$.

When the state space $\mathbb{S} = \{1, 2, \ldots\}$, the above construction goes over with minor notational modifications.

Next consider the case $\mathbb{S} = \mathbb{R}$. A function $Q : \mathbb{R} \times \mathcal{B}(\mathbb{R}) \to [0, 1]$ is called a *probability transition function* if

(i) for each x in $\mathbb{R}$, $Q(x, \cdot)$ is a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and

(ii) for each B in $\mathcal{B}(\mathbb{R})$, $Q(\cdot, B)$ is a Borel measurable function on $\mathbb{R}$.

Let μ be a probability distribution on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Using Theorem 6.3.4, it can be shown that there exists a stochastic process $\{X_n\}_{n \geq 1}$ such that

$$\begin{aligned} &P(X_1 \in B_1, X_2 \in B_2, \ldots, X_n \in B_n) \\ &= \int_{B_1} \int_{B_2} \cdots \int_{B_{n-1}} Q(x_{n-1}, B_n) Q(x_{n-2}, dx_{n-1}) \cdots Q(x_1, dx_2) \mu(dx_1), \end{aligned} \tag{3.16}$$

where right side of (3.16) is a well-defined probability measure on $\big(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n)\big)$ (Problem 6.18). Such a sequence $\{X_n\}_{n \geq}$ is called a Markov chain with *state space* $\mathbb{R}$, *initial distribution* μ, and *transition probability function* Q. For more on Markov chains, see Chapter 14.

Example 6.3.9: (*Gaussian processes*). Let A be a nonempty set and $\{X_\alpha : \alpha \in A\}$ be a stochastic process. Such a process is called *Gaussian* if for $\{\alpha_1, \alpha_2, \ldots, \alpha_k\} \subset A$ and real numbers $t_1, t_2, \ldots, t_k$, the random variable $\sum_{i=1}^k t_i X_{\alpha_i}$ has a univariate normal distribution (with possibly zero variance). For such a process, the functions $\mu(\alpha) \equiv EX_\alpha$ and $\sigma(\alpha, \beta) \equiv$

$\mathrm{Cov}(X_\alpha, X_\beta)$ are called the *mean and covariance functions*, respectively. Since $\mathrm{Var}(\sum_{i=1}^k t_i X_{\alpha_i}) \geq 0$, it follows that for any $t_1, t_2, \ldots, t_k$,

$$\sum_{i=1}^k \sum_{j=1}^k t_i t_j \sigma(\alpha_i, \alpha_j) \geq 0. \tag{3.17}$$

This property of the covariance function $\sigma(\cdot, \cdot)$ is called *nonnegative definiteness.*

A natural question is: Given functions $\mu : A \to \mathbb{R}$ and $\sigma : A \times A \to \mathbb{R}$ such that σ is symmetric and satisfies (3.17), does there exist a Gaussian process $\{X_\alpha : \alpha \in A\}$ with $\mu(\cdot)$ and $\sigma(\cdot ;)$ as its mean and covariance functions, respectively? The answer is yes and it follows from Theorem 6.3.1 by defining the family Q_A of finite dimensional distributions as follows. Let $\nu_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}$ be the unique probability distribution on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ with the moment generating function

$$\begin{aligned} & M_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}(s_1, s_2, \ldots, s_k) \\ = \quad & \exp\left(\sum_{i=1}^k s_i \mu(\alpha_i) + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k s_i s_j \sigma(\alpha_i, \alpha_j)\right) \end{aligned} \tag{3.18}$$

for $s_1, s_2, \ldots, s_k$ in $\mathbb{R}$. If the matrix $\Sigma \equiv \big((\sigma(\alpha_i, \alpha_j))\big)$, $1 \leq i, j \leq k$ is *positive definite*, i.e., it is such that in (3.17) equality holds iff $t_i = 0$ for all i, then $\nu_{(\alpha_1, \ldots, \alpha_k)}(\cdot)$ can be shown to be a probability measure that is absolutely continuous w.r.t. m^k, the Lebesgue measure on $\mathbb{R}^k$ with density $\frac{1}{(2\pi)^{k/2}} |\Sigma|^{-\frac{1}{2}} e^{-\sum_{i=1}^k \sum_{j=1}^k \left(x_i - \mu(\alpha_i)\right) \tilde{\sigma}_{ij} \left(x_j - \mu(\alpha_j)\right)/2}$ where $\tilde{\Sigma} \equiv \big((\tilde{\sigma}_{ij})\big) = \Sigma^{-1}$, the inverse of Σ and $|\Sigma| =$ the determinant of Σ.

The verification of conditions (3.1) and (3.2) for this family is left as an exercise (Problem 6.12).

Remark 6.3.4: Kolmogorov's consistency theorem (Theorem 6.3.1) remains valid when the real line $\mathbb{R}$ is replaced by a complete separable metric space $\mathbb{S}$. More specifically, let A be a nonempty set and for $\{\alpha_1, \alpha_2, \ldots, \alpha_k\} \subset A$, $1 \leq k < \infty$, let $\nu_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}(\cdot)$ be a probability measure on $(\mathbb{S}^k, \mathcal{B}(\mathbb{S}^k))$. If the family $Q_A \equiv \{\nu_{(\alpha_1, \alpha_2, \ldots, \alpha_k)} : \{\alpha_1, \alpha_2, \ldots, \alpha_k\} \subset A,\ 1 \leq k < \infty\}$ satisfies the natural analogs of (3.1) and (3.2), then there exists a probability measure P on $(\Omega \equiv \mathbb{S}^A, \mathcal{F} \equiv (\mathcal{B}(\mathbb{S}))^A)$ and an $\mathbb{S}$-valued stochastic process $\{X_\alpha : \alpha \in A\}$ on $(\Omega, \mathcal{F}, P)$ such that $\nu_{(\alpha_1, \alpha_2, \ldots, \alpha_k)}(\cdot) = P(X_{\alpha_1}, X_{\alpha_2}, \ldots X_{\alpha_K})^{-1}(\cdot)$. Here $\mathbb{S}^A$ is the set of all $\mathbb{S}$ valued functions on A, $(\mathcal{B}(\mathbb{S}))^A$ is the σ-algebra generated by the cylinder sets of the form

$$C = \{f : f : A \to \mathbb{S},\ f(\alpha_i) \in B_i,\ i = 1, 2, \ldots, k\}$$

where $\{\alpha_1, \alpha_2, \ldots, \alpha_k\} \subset A$, $B_i \in \mathcal{B}(\mathbb{S})$, $1 \leq i \leq k$, $1 \leq k < \infty$, and also $X_\alpha(\omega)$ is the projection map $X_\alpha(\omega) \equiv \omega(\alpha)$. The main step in the proof of Theorem 6.3.1 was to establish the countable additivity of the set function P on the algebra $\mathcal{C}$ of finite dimensional cylinder sets. This in turn depended upon the fact that any probability measure μ on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ for $1 \leq k < \infty$ is *regular*, i.e., for every Borel set B in $\mathcal{B}(\mathbb{R}^k)$ and for every $\epsilon > 0$, there exists a compact set $G \subset B$ such that $\mu(B \backslash G) < \epsilon$. If $\mathbb{S}$ is a Polish space, then any probability measure on $(\mathbb{S}^k, (\mathcal{B}(\mathbb{S}))^k)$, $1 \leq k < \infty$ is regular (see Billingsley (1968)), and hence, the main steps in the proof of Theorem 6.3.1 go through in this case.

Remark 6.3.5: (*Limitations of Theorem 6.3.1*). In this construction, $\Omega = \mathbb{R}^A$ is rather large and the σ-algebra $\mathcal{F} \equiv (\mathcal{B}(\mathbb{R}))^A$ is not large enough to include many events of interest when the index set A is uncountable. In fact, it can be shown that $\mathcal{F}$ coincides with the class of all sets $G \subset \Omega$ that depend only on a countable number of coordinates of ω. More precisely, the following holds.

Proposition 6.3.5: *The σ-algebra*

$$\begin{aligned} \mathcal{F} &= \{G : G = \pi_{A_1}^{-1}(B) \textit{ for some } B \textit{ in } \mathcal{B}(\mathbb{R}^\infty) \\ &\qquad \textit{and} \quad A_1 \subset A, \ A_1 \quad \textit{countable}\}. \end{aligned} \tag{3.19}$$

Proof: Verify that the right side of (3.19) is a σ-algebra containing the class $\mathcal{C}$ of cylinder sets and also that, it is contained in $\mathcal{F}$. □

For example, if $A = [0, 1]$, then the set $C[0, 1]$ of all continuous functions from $[0, 1] \to \mathbb{R}$ is not a member of $\mathcal{F} \equiv (\mathcal{B}(\mathbb{R}))^A$. Similarly, if $M(\omega) \equiv \sup\{|\omega(\alpha)| : \alpha \in [0, 1]\}$, then the set $\{M(\omega) \leq 1\}$ is not in $\mathcal{F} = (\mathcal{B}(\mathbb{R}))^{[0,1]}$. When A is an interval in $\mathbb{R}$, this difficulty can be overcome in several ways. One approach pioneered by J.L. Doob is the notion of separable stochastic processes (Doob (1953)). Another approach pioneered by Kolmogorov and Skorohod is to restrict Ω to the class of all continuous functions or functions that are right continuous and have left limits (Billingsley (1968)). For more on stochastic processes, see Chapter 15.

Independent Random Experiments

If $\mathcal{E}_1$ and $\mathcal{E}_2$ are two random experiments with associated probability spaces $(\Omega_1, \mathcal{F}_1, P_1)$ and $(\Omega_2, \mathcal{F}_2, P_2)$, it is possible to model the experiment of performing both $\mathcal{E}_1$ and $\mathcal{E}_2$ independently by the product probability space $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2, P_1 \times P_2)$ (see Chapter 5). The same idea carries over to an arbitrary collection $\{\mathcal{E}_\alpha : \alpha \in A\}$ of random experiments. It is possible to think of a grand experiment $\mathcal{E}$ in which all the $\mathcal{E}_\alpha$'s are independent components by considering the product probability space

$$(\times_{\alpha \in A} \Omega_\alpha, \times_{\alpha \in A} \mathcal{F}_\alpha, \times_{\alpha \in A} P_\alpha) \equiv (\Omega, \mathcal{F}, P) \tag{3.20}$$

where $(\Omega_\alpha, \mathcal{F}_\alpha, P_\alpha)$ is the probability space corresponding to $\mathcal{E}_\alpha$. Here $\Omega \equiv \times_{\alpha \in A} \Omega_\alpha$ is the collection of all functions ω on A such that $\omega(\alpha) \in \Omega_\alpha$,

$\mathcal{F} \equiv \times_{\alpha \in A} \mathcal{F}_\alpha$ is the σ-algebra generated by finite dimensional cylinder sets of the form

$$C = \{\omega : \omega(\alpha_i) \in B_{\alpha_i},\ i = 1, 2, \ldots, k\}, \tag{3.21}$$

$1 \leq k < \infty, \{\alpha_1, \alpha_2, \ldots, \alpha_k\} \subset A$, $B_{\alpha_i} \in \mathcal{F}_{\alpha_i}$ and $P \equiv \times_{\alpha \in A} P_\alpha$ is the probability measure on $\mathcal{F}$ such that for C of (3.21),

$$P(C) = \prod_{i=1}^{k} P_{\alpha_i}(B_{\alpha_i}). \tag{3.22}$$

The proof of the existence of such a P on $\mathcal{F}$ is an application of the extension theorem (Theorem 1.3.3). The verification of countable additivity on the class $\mathcal{C}$ of cylinder sets is not difficult. See Kolmogorov (1956).

6.4 Problems

6.1 Let $\mu_1 = \mu_2$ be the probability distribution on $\Omega = \{1, 2\}$ with $\mu_1(\{1\}) = 1/2$. Find two distinct probability distributions on $\Omega \times \Omega$ with μ_1 and μ_2 as the set of marginals.

6.2 Let $\Omega = (0, 1)$, $\mathcal{F} = \mathcal{B}((0, 1))$ and P be the Lebesgue measure on (0,1). Let $X(\omega) = -\log \omega$, $h(x) = x^2$ and $Y = h(X)$. Find P_X and P_Y and evaluate EY by applying the change of variables formula (Proposition 6.2.1).

6.3 In the change of variables formula, one of the three integrals is usually easier to evaluate than the other two. In this problem, in part (a), the first integral is easier to evaluate than the other two while in part (b), the second one is easier.

(a) Let $Z \sim N(0, 1)$, $X = Z^2$, and $Y = e^{-X}$.

(i) Find the distributions P_X and P_Y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

(ii) Compute the integrals

$$\int_{\mathbb{R}} e^{-z^2} \phi(z) dz, \quad \int_{\mathbb{R}} e^{-x} P_X(dx) \quad \text{and} \quad \int_{\mathbb{R}} y P_Y(dy),$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, $-\infty < z < \infty$. Verify that all three integrals agree.

(b) Let $X_1, X_2, \ldots, X_n$ be iid $N(0, 1)$ random variables. Let $Y = (X_1 + \cdots + X_k)$ and $Z = Y^2$.

(i) Find the distributions of Y and Z.

(ii) Evaluate $\int_{\mathbb{R}^k} (x_1 + \cdots + x_k)^2 dP_{X_1, \ldots, X_k}(x_1, \ldots, x_k)$, $\int_{\mathbb{R}} y^2 P_Y(dy)$, and $\int_{\mathbb{R}_+} z P_Z(dz)$.

(c) Let $X_1, X_2, \ldots, X_k$ be independent Binomial (n_i, p), $i = 1, 2, \ldots, k$ random variables. Let $Y = (X_1 + \cdots + X_k)$.

  (i) Find the distribution P_Y of Y.

  (ii) Evaluate $\int_{\mathbb{R}^k} (x_1 + \cdots + x_k) dP_{X_1,\ldots,X_k}(x_1, \ldots, x_k)$ and $\int_{\mathbb{R}} y P_Y(dy)$.

6.4 Let X be a random variable such that $M_X(t) \equiv E(e^{tX}) < \infty$ for $|t| < \epsilon$ for some $\epsilon > 0$.

(a) Show that $E(e^{tX}|X|^r) < \infty$ for all $r > 0$ and $|t| < \epsilon$.

(b) Show that $M_X^{(r)}(t)$, the rth derivative of $M_X(t)$ for $r \in \mathbb{N}$, satisfies

$$M_X^{(r)}(t) = E(e^{tX} X^r) \quad \text{for} \quad |t| < \epsilon.$$

(c) Verify (2.25).

(**Hint:** (a) First show that for $t_1 \in (-\epsilon, \epsilon)$, there exist a $t_2 \in (-\epsilon, \epsilon)$ such that $|t_1| < |t_2| < \epsilon$ and for some $C < \infty$, $e^{t_1 x}|x|^r \leq C e^{|t_2 x|}$ for all x in $\mathbb{R}$.
(b) Verify that for all $x \in \mathbb{R}$, $|e^x - 1| \leq |x| e^{|x|}$. Now use (a) and the DCT to show that $M_X(t)$ is differentiable and $M_X^{(1)}(t) = E(e^{tX} X)$ for all $|t| < \epsilon$. Now complete the proof by induction.)

6.5 Let X be a random variable.

(a) Show that $\phi(r) \equiv (E|X|^r)^{1/r}$ is nondecreasing on $(0, \infty)$.

(b) Show that $\phi(r) \equiv \log E|X|^r$ is convex in $(0, r_0)$ if $E|X|^{r_0} < \infty$.

(c) Let $M = \sup\{x : P(|X| > x) > 0\}$. Show that

  (i) $\lim\limits_{r \uparrow \infty} \phi(r) = M$.

  (ii) $\lim\limits_{n \to \infty} \frac{E|X|^{n+1}}{E|X|^n} = M$.

(**Hint:** For $M < \infty$, note that $E|X|^r \geq (M - \epsilon)^r P(|X| > M - \epsilon)$ for any $\epsilon > 0$.)

6.6 Show that if equality holds in (2.20), then there exist constants a and b such that $P(Y = aX + b) = 1$.

(**Hint:** Show that there exist a constant a such that $\mathrm{Var}(Y - aX) = 0$.)

6.7 Determine C and its base B explicitly in Examples 6.3.3 and 6.3.4.

6.8 (a) Show that D in Example 6.3.5 is not a finite dimensional cylinder set.

(**Hint:** Note that $\lim\limits_{n \to \infty} \frac{1}{n} \sum_{j=1}^{n} x_j$ is not determined by the values of finitely many x_i's.)

(b) Find three other such examples of sets D in $\mathbb{R}^\infty$ that are not finite dimensional cylinder sets.

6.9 Establish the assertion in Remark 6.3.3 by completing the following steps:

(a) Show that the coordinate map $f_n(x) \equiv x_n$ from $\mathbb{R}^\infty$ to $\mathbb{R}$ is continuous under the metric d of (3.3). (Conclude, using Example 1.1.6, that $\mathcal{R}^{\mathbb{N}} \subset \mathcal{B}(\mathbb{R}^\infty)$).

(b) Let $\mathcal{C}_1 \equiv \{A : A = (a_1, b_1) \times \cdots \times (a_k, b_k) \times \mathbb{R}^\infty,\ -\infty \le a_i < b_i \le \infty,\ 1 \le i \le k$, for some $k < \infty\}$ and $\mathcal{C}_2 \equiv \{A : A$ is an open ball in $(\mathbb{R}^\infty, d)\}$. Show that $\sigma\langle \mathcal{C}_2 \rangle \subset \sigma\langle \mathcal{C}_1 \rangle$.

(c) Show that $\sigma\langle \mathcal{C}_2 \rangle = \mathcal{B}(\mathbb{R}^\infty)$ by showing that every open set in $(\mathbb{R}^\infty, d)$ is a countable union of open balls.

6.10 Show that the family $Q_{\mathbb{N}}$ defined in (3.12) satisfies the consistency conditions (3.1) and (3.2) of Theorem 6.3.1.

6.11 Verify that the family of finite dimensional distributions defined by the right side of (3.14) satisfies the conditions of Theorem 6.3.4.

6.12 Verify that the family of distributions defined in (3.18) satisfies conditions (3.1) and (3.2) of Theorem 6.3.1.

(**Hint:** Use the fact that for any $k \ge 1$, any $\mu = (\mu_1, \mu_2, \ldots, \mu_k) \in \mathbb{R}^k$, and any nonnegative definite $k \times k$ matrix $\Sigma \equiv ((\sigma_{ij}))_{k \times k}$, there is a unique probability distribution ν such that for any $s = (s_1, s_2, \ldots, s_k)$ in $\mathbb{R}^k$,

$$\begin{aligned} &\int_{\mathbb{R}^k} \exp\left(\sum_{i=1}^k s_i x_k\right) \nu(dx) \\ = \quad & \exp\left(\sum_{i=1}^k s_i \mu_i + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k s_i s_j \sigma_{ij}\right). \end{aligned}$$

Observe that this implies that for $s = (s_1, s_2, \ldots, s_k)$ in $\mathbb{R}^k$, the induced distribution (under ν) on $\mathbb{R}$ by the map $g(x) = \sum_{i=1}^k s_i x_i$ from $\mathbb{R}^k \to \mathbb{R}$ is univariate normal with mean $\sum_{i=1}^k s_i \mu_i$ and variance $\sum_{i=1}^k \sum_{j=1}^k s_i s_j \sigma_{ij}$.)

6.13 Show that the set $D \equiv C[0,1]$ of continuous functions from $[0,1]$ to $\mathbb{R}$ is not a member of the σ-algebra $\mathcal{F} \equiv (\mathcal{B}(\mathbb{R}))^{[0,1]}$.

(**Hint:** If $D \in \mathcal{F}$, then by Proposition 6.3.5, D is of the form $\pi_{A_1}^{-1}(B)$ for some B in $\mathcal{B}(\mathbb{R}^\infty)$, where $A_1 \subset [0,1]$ is countable. Show that for any such A_1 and B, there exist functions $f : [0,1] \to \mathbb{R}$ such that $f \in \pi_{A_1}^{-1}(B)$ but f is not continuous on $[0,1]$.)

6.14 Show that $K \equiv \big\{\omega : \omega \in \mathbb{R}^{[0,1]},\ \sup_{0\le\alpha\le 1} |\omega(\alpha)| < 1\big\}$ is not in $\mathcal{F} \equiv (\mathcal{B}(\mathbb{R}))^{[0,1]}$.

(**Hint:** Observe that $\sup_{0\le\alpha\le 1} |\omega(\alpha)|$ is not determined by the values of $\omega(\alpha)$ for countably many α's.)

6.15 Let $\{\mu_i\}_{i\ge 1}$ be a sequence of probability distributions on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$ and let μ be a probability distribution on $\mathbb{N}$ with $p_i \equiv \mu(\{i\})$, $i \ge 1$.

(a) Verify that $\nu(\cdot) \equiv \sum_{i\ge 1} p_i \mu_i(\cdot)$ is a probability distribution on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$.

(b) (i) Show that there exists a probability space $(\Omega, \mathcal{F}, P)$ and a collection of independent random variables $\{J, X_1, X_2, \ldots\}$ on $(\Omega, \mathcal{F}, P)$ such that for each $i \ge 1$, X_i has distribution μ_i and $J \sim \mu$.

(ii) Let $Y = X_J$, i.e., $Y(\omega) \equiv X_{J(\omega)}(\omega)$. Show that Y is a random variable on $(\Omega, \mathcal{F}, P)$ and $Y \sim \nu$.

6.16 Let F be a cdf on $\mathbb{R}$ and let F be decomposed as

$$F = \alpha F_d + \beta F_{ac} + \gamma F_{sc}$$

where $\alpha, \beta, \gamma \in [0, 1]$ and $\alpha + \beta + \gamma = 1$ and F_d, F_{ac}, F_{sc} are discrete, absolutely continuous, and singular continuous cdfs on $\mathbb{R}$ (cf. (4.5.3)). Show that there exist independent random variables X_1, X_2, X_3 and J on some probability space such that $X_1 \sim F_d$, $X_2 \sim F_{ac}$, $X_3 \sim F_{sc}$, $P(J = 1) = \alpha$, $P(J = 2) = \beta$, $P(J = 3) = \gamma$ and $X_J \sim F$, where $\sim$ means "has cdf".

6.17 Let μ be a probability measure on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$. Let for each x in $\mathbb{R}$, $F(x, \cdot)$ be a cdf on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$. Let $\psi(x, t) \equiv \inf\{y : F(x, y) \ge t\}$, for x in $\mathbb{R}$, $0 < t < 1$. Assume that $\psi(\cdot, \cdot) : \mathbb{R} \times (0, 1) \to \mathbb{R}$ is measurable. Let X and U be independent random variables on some probability space $(\Omega, \mathcal{F}, P)$ such that $X \sim \mu$ and $U \sim$ uniform (0,1).

(a) Show that $Y = \psi(X, U)$ is a random variable.

(b) Show that $P(Y \le y) = \int_{\mathbb{R}} F(x, y)\mu(dx)$. (The distribution of Y is called a mixture of distributions with μ as the mixing distribution. This is of relevance in Bayesian statistical inference.)

6.18 (a) Let $(\mathbb{S}_i, \mathcal{S}_i)$, $i = 1, 2$ be two measurable spaces. Let μ be a probability measure on $(\mathbb{S}_1, \mathcal{S}_1)$ and let $Q : \mathbb{S}_1 \times \mathcal{S}_2 \to [0, 1]$ be such that for each x in $\mathbb{S}_1$, $Q(x, \cdot)$ is a probability measure on $(\mathbb{S}_2, \mathcal{S}_2)$ and for each B in $\mathcal{S}_2$, $Q(\cdot, B)$ is $\mathcal{S}_1$-measurable. Define

$$\nu(B_1 \times B_2) \equiv \int_{B_1} Q(x, B_2)\mu(dx)$$

on $\mathcal{C} \equiv \{B_1 \times B_2 : B_i \in s_i,\ i = 1, 2\}$. Show that ν can be extended to be a probability measure on $\sigma\langle\mathcal{C}\rangle \equiv \mathcal{S}_1 \times \mathcal{S}_2$.

(b) Let μ and Q be as in Example 6.3.8 (cf. (3.16)). For each $n \geq 1$ let ν_n be a set function defined by the recursive scheme

$$\begin{aligned}\nu_1(\cdot) &= \mu(\cdot),\\ \nu_{n+1}(A \times B) &= \int_A Q(x, B)\nu_n(dx),\ A \in \mathcal{B}(\mathbb{R}^n),\ B \in \mathcal{B}(\mathbb{R}).\end{aligned}$$

Show that for each n, ν_n can be extended to be a probability measure on $\big(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n)\big)$. (Thus the right side of (3.16) is defined to be $\nu_n(B_1 \times B_2 \times \cdots \times B_n)$.)

6.19 (*Bayesian paradigm*). Consider the setup in Problem 6.18 (a). Let $\lambda(B) \equiv \nu(\mathbb{S}_1 \times B) = \int_{\mathbb{S}_1} Q(x, B)\mu(dx)$ for all B in $\mathcal{S}_2$.

(a) Verify that λ is a probability measure on $(\mathbb{S}_2, \mathcal{S}_2)$.

(b) Now fix B_1 in $\mathcal{S}_1$. Show that there exists a function $\tilde{Q}(x, B_1)$, $\mathbb{S}_2 \to [0, 1]$ that is $\langle\mathcal{S}_2, \mathcal{B}(\mathbb{R})\rangle$-measurable such that

$$\nu(B_1 \times B_2) = \int_{B_2} \tilde{Q}(x, B_1)\lambda(dx).$$

(**Hint:** Apply the Radon-Nikodym theorem to the pair $\nu(B_1 \times \cdot)$ and $\lambda(\cdot)$.)

(c) Let $\Omega = \mathbb{S}_1 \times \mathbb{S}_2$, $\mathcal{F} = \sigma\langle\mathcal{C}\rangle$. For $\omega = (s_1, s_2)$, let $\theta(\omega) = s_1$ and $X(\omega) = s_2$. Think of θ as the *parameter*, X as the *data*, $Q(\theta, \cdot)$ as the *distribution of* X *given* θ, $\mu(\cdot)$ as the *prior distribution of* θ and $\tilde{Q}(x, B_1)$ as the *posterior* probability that θ is in B_1 given the data $X = x$. Compute $\tilde{Q}(x, B_1)$ when $(\mathbb{S}_i, \mathcal{S}_i) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $i = 1, 2$, $\mu(\cdot) \sim N(0, 1)$, $Q(\theta, \cdot) \sim N(\theta, 1)$.

6.20 Let X be a random variable on some probability space $(\Omega, \mathcal{F}, P)$. Recall that a random variable X is

(a) *discrete* if there is a finite or countable set $D \equiv \{a_j : 1 \leq j \leq k \leq \infty\}$ such that $P(X \in D) = 1$,

(b) *continuous* if for every $x \in \mathbb{R}$, $P(X = x) = 0$ or equivalently the cdf $F_X(\cdot)$ is continuous on all of $\mathbb{R}$,

(c) *absolutely continuous* if there exists a nonnegative Borel measurable function $f_X(\cdot)$ on $\mathbb{R}$ such that for any $-\infty < a < b < \infty$,

$$P(a < X \leq b) = \int_{(a,b]} f_X(\cdot)dm$$

or equivalently the induced measure PX^{-1} is $\ll m$,

(d) *singular* if $PX^{-1} \perp m$ or equivalently $F_X(\cdot) = 0$ a.e. m,

(e) *singular continuous* if it is singular and continuous.

Let $g : \mathbb{R} \to \mathbb{R}$ be Borel measurable and $Y = g(X)$.

(a) Show that if X is discrete then so is Y but not conversely.

(b) Show that if X is continuous and g is (1–1) on the range of X, then Y is continuous.

(c) Show if X is absolutely continuous with pdf $f_X(\cdot)$ and g is absolutely continuous on bounded intervals such that $g'(\cdot) > 0$ a.e. (m), then Y is also absolutely continuous with pdf

$$f_Y(y) = \frac{f_X\big(g^{-1}(y)\big)}{g'\big(g^{-1}(y)\big)}.$$

(d) Let X be as in (c) above. Suppose g is absolutely continuous on bounded intervals and there exist disjoint intervals $\{I_j\}_{1 \le j \le k}$, $1 \le k \le \infty$, such that $\bigcup_{1 \le j \le k} I_j = \mathbb{R}$ and for each j, either $g'(\cdot) > 0$ a.e. (m) on I_j or $g'(\cdot) < 0$ a.e. (m) on I_j. Show that Y is also absolutely continuous with pdf

$$f_Y(y) = \sum_{x_j \in D(y)} \frac{f_X(x_j)}{|g'(x_j)|}$$

where $D\{y\} \equiv \{x_j : x_j \in I_j,\, g(x_j) = y\}$.

(e) Use (c) to compute the pdf of Y when

(i) $X \sim N(0,1)$, $g(x) = e^x$.

(ii) $X \sim N(0,1)$, $g(x) = x^2$.

(iii) $X \sim N(0,1)$, $g(x) = \sin 2\pi x$.

(iv) $X \sim \exp(1)$, $g(x) = e^{-x}$.

6.21 (*Simple random sampling without replacement*). Let $S \equiv \{1, 2, \ldots, m\}$, $1 < m < \infty$. Fix $1 \le n \le m$. Choose an element X_1 from S such that the probability that $X_1 = j$ is $\frac{1}{m}$ for all $j \in S$. Next, choose an element X_2 from $S - \{X_1\}$ such that the probability that $X_2 = j$ is $\frac{1}{(m-1)}$ for $j \in S - \{X_1\}$. Continue this procedure for n steps. Write the outcome as the ordered vector $\omega \equiv (X_1, X_2, \ldots, X_n)$.

(a) Identify the sample space Ω, the σ-algebra $\mathcal{F}$ and the probability measure P for this experiment.

(b) Show that for any permutation σ of $\{1, 2, \ldots, n\}$, the random vector $Y_\sigma = (X_{\sigma(1)}, X_{\sigma(2)}, \ldots, X_{\sigma(n)})$ has the same distribution as $(X_1, X_2, \ldots, X_n)$.

(c) Conclude that $\{X_i\}_{1 \le i \le n}$ are identically distributed and that EX_i, $\mathrm{Cov}(X_i, X_j)$, $i \neq j$ are independent of i and j and compute them.

(d) Answer the same questions (a)–(c) if the sampling is changed to *with replacement*, i.e., at each stage i, the probability that $P(X_i = j) = \frac{1}{m}$ for all $j \in S$.

(e) In (d), let D be the number of *distinct units* in the sample. Find $E(D)$ and $\mathrm{Var}(D)$.

6.22 Let X be a nonnegative random variable. Show that

$$\sqrt{1 + (EX)^2} \le E\sqrt{1 + X^2} \le 1 + EX.$$

(Note that $f(x) \equiv \sqrt{1 + x^2}$ is convex on $[0, \infty)$ and bounded by $1+x$.)

6.23 Let X and Y be nonnegative random variables defined on a probability space $(\Omega, \mathcal{F}, P)$. Suppose $X \cdot Y \ge 1$ w.p. 1. Show that

$$EX \cdot EY \ge 1.$$

(**Hint:** Use Cauchy-Schwarz on $\sqrt{X}\sqrt{Y}$.)

6.24 Let μ be a probability measure on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$. Show that there is a random variable X on the Lebesgue space $([0,1], \mathcal{B}([0,1]), m)$ such that $m\,X^{-1} \equiv \mu$ where m is the Lebesgue measure. Extend this to $\big(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k)\big)$, where k is an integer > 1.

(Note: This is true for any Polish space, i.e., a complete separable metric space, see Billingsley (1968).)